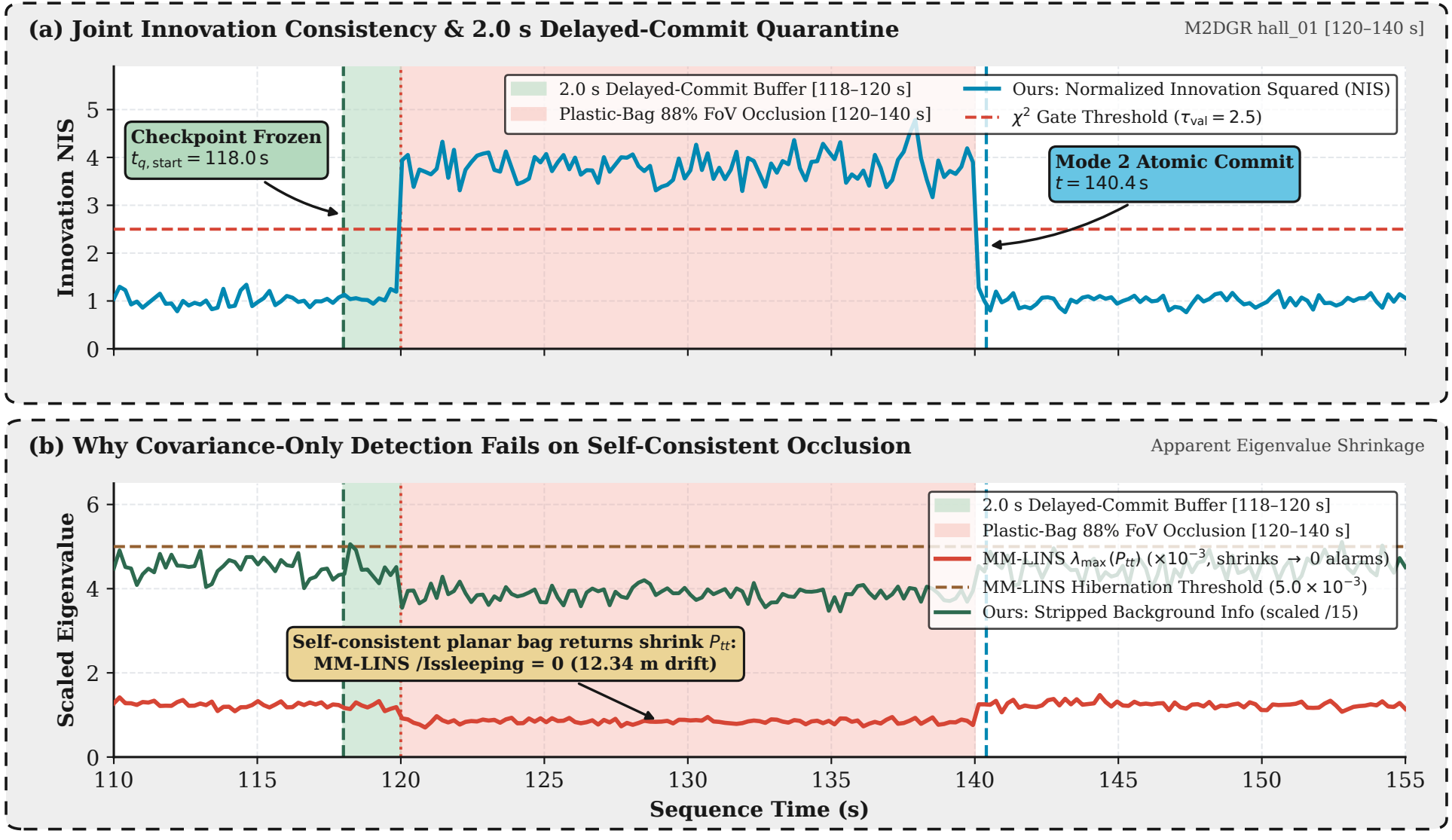

(a) Joint Innovation Consistency & 2.0 s Delayed-Commit Quarantine
M2DGR hall_01 [120–140 s]
Innovation NIS
2.0 s Delayed-Commit Buffer [118–120 s]
Plastic-Bag 88% FoV Occlusion [120–140 s]
Ours: Normalized Innovation Squared (NIS)
$\chi^2$ Gate Threshold ($\tau_{val} = 2.5$)
Checkpoint Frozen
$t_{q,\,start} = 118.0$ s
Mode 2 Atomic Commit
$t = 140.4$ s
(b) Why Covariance-Only Detection Fails on Self-Consistent Occlusion
Apparent Eigenvalue Shrinkage
Scaled Eigenvalue
2.0 s Delayed-Commit Buffer [118–120 s]
Plastic-Bag 88% FoV Occlusion [120–140 s]
MM-LINS $\lambda_{max}(P_{tt})$ ($\times 10^{-3}$, shrinks → 0 alarms)
MM-LINS Hibernation Threshold ($5.0 \times 10^{-3}$)
Ours: Stripped Background Info (scaled /15)
Self-consistent planar bag returns shrink $P_{tt}$:
MM-LINS /Issleeping = 0 (12.34 m drift)
Sequence Time (s)